Build your cardboard viewer

ScratchXR | English guide + full-size Cardboard 1.0 template

A hands-on activity to explore stereo viewing on a phone. This is an older template: first check the fit with a paper mock-up. Large phones may not fit; do not enlarge the drawing to accommodate them.

Gather your materials

- 6 A4 sheets for the template, printed on one side only.
- Thin, firm cardboard, about 1.5 mm thick; approximately 22 x 56 cm minimum.
- Two Cardboard v1 lenses matching the holes and lens-to-screen spacing. Google's reference suggests a focal length of about 45 mm.
- A ruler, pencil, cardboard-safe glue, tape, hook-and-loop fastener and a rubber band.
- Scissors; an adult should make any internal cuts requiring a craft knife.

Keep the original dimensions

Print pages 3 to 8 on A4 paper, portrait orientation, actual size / 100%. Turn off 'fit to page' and multiple-pages-per-sheet printing.

Measure the 50 mm ruler on every sheet. If it is not exactly 5 cm, correct the print settings before cutting.

Each original sheet is split into two halves with a 10 mm overlap. Cut and fold lines retain the dimensions of the source PDF.

Cardboard cannot replace the lenses. Website viewing also depends on the phone, sensors and browser. A physical assembly and phone-fit test are still required.

From paper to viewer

Read these steps before cutting the cardboard.

1. Join the sheets

Join A-left to A-right, then repeat for B and C. Trim only the white margin at the seam and overlap the drawings. Align the upper and lower crosses and the template lines. The repeated strip is 10 mm wide.

2. Prepare the parts

Join the body sections using matching numbers, light numbers over dark ones. Check your phone's fit in paper first. Glue the template to cardboard. Black lines are cuts; red lines are folds. Do not cut the fold lines.

3. Assemble the support

Place the lenses in part A, following the illustrated orientation. Fold the three lens-holder panels to retain the lenses. Fit the central divider and body tabs without forcing or distorting the lenses.

4. Secure and try it

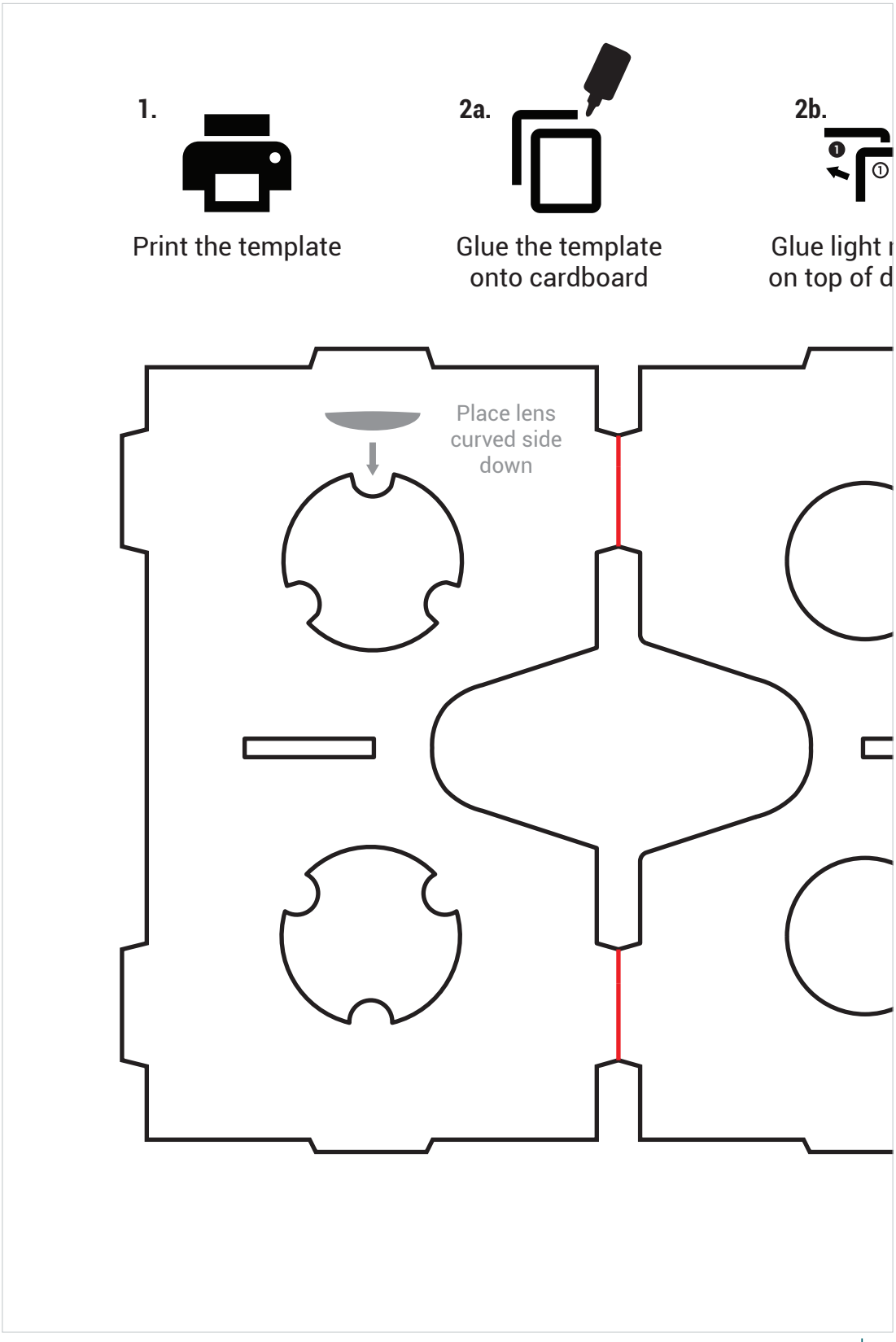
Secure the closure with hook-and-loop fastener and retain the phone with a rubber band. Check it cannot slide out. Open the stereo experience before inserting the phone. Hold the viewer while seated; remove it if you see double or feel discomfort.

The original drawing includes magnets and a magnetic button. They are not needed to support the viewer, and browser input compatibility is not guaranteed. Keep small parts under adult supervision. Keep the lenses out of direct sunlight.

Source: Cardboard 1.0 template preserved by ScienceAll; Google material guidelines. This adaptation adds instructions and splits the sheets into A4 pages. It is not viewer certification or a guarantee of compatibility with every phone.

Template A - left

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



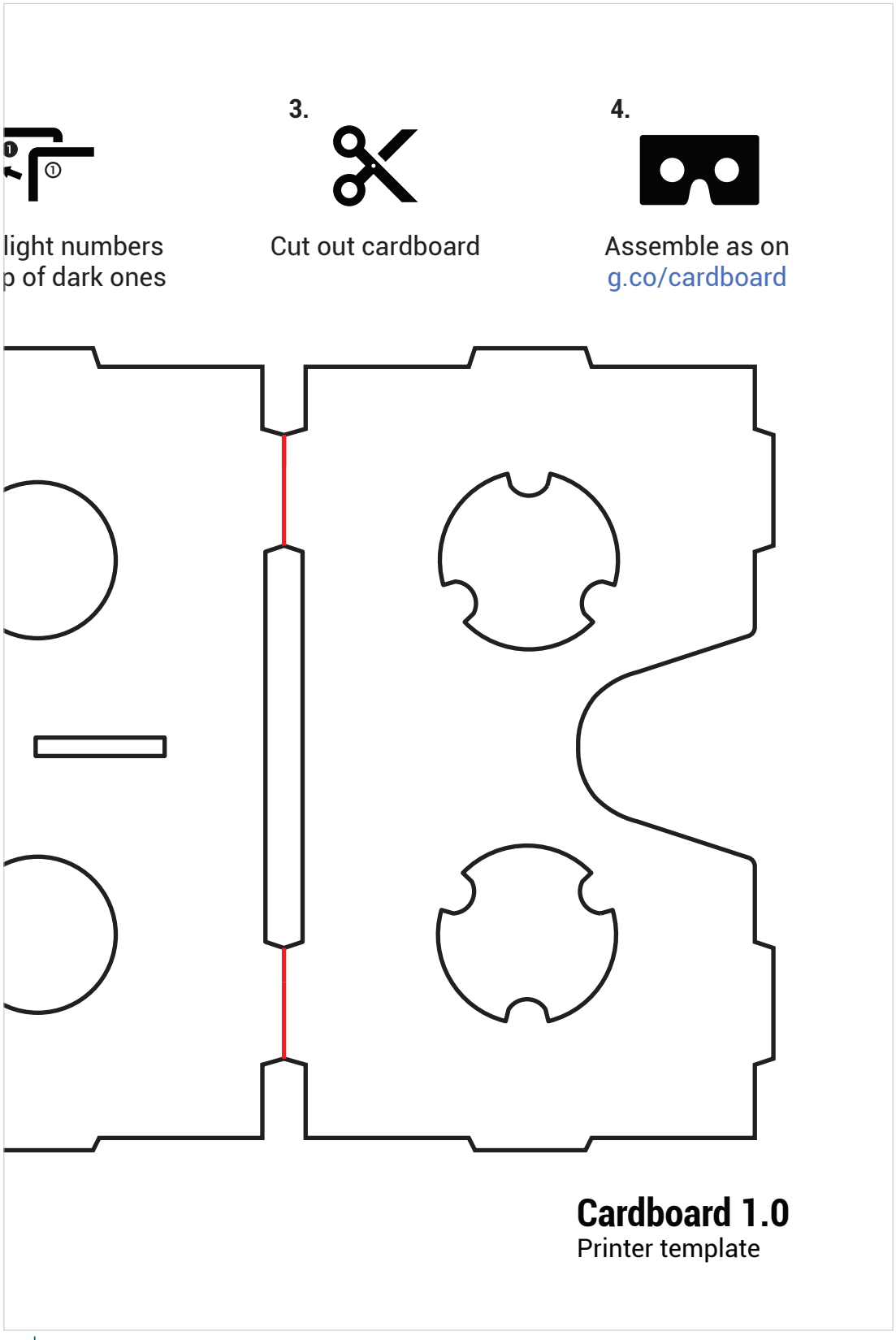
Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap

Template A - right

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



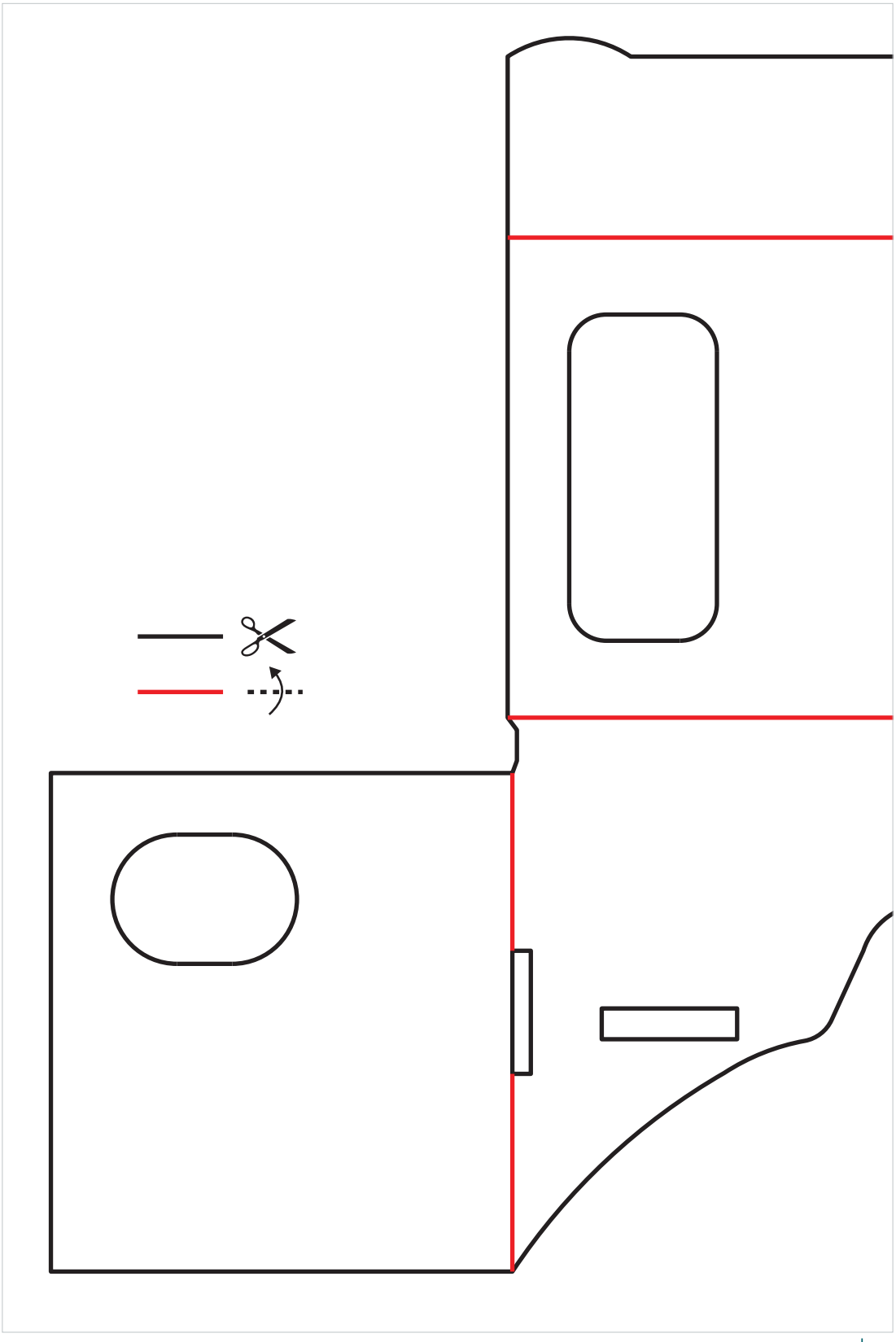
Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap

Template B - left

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



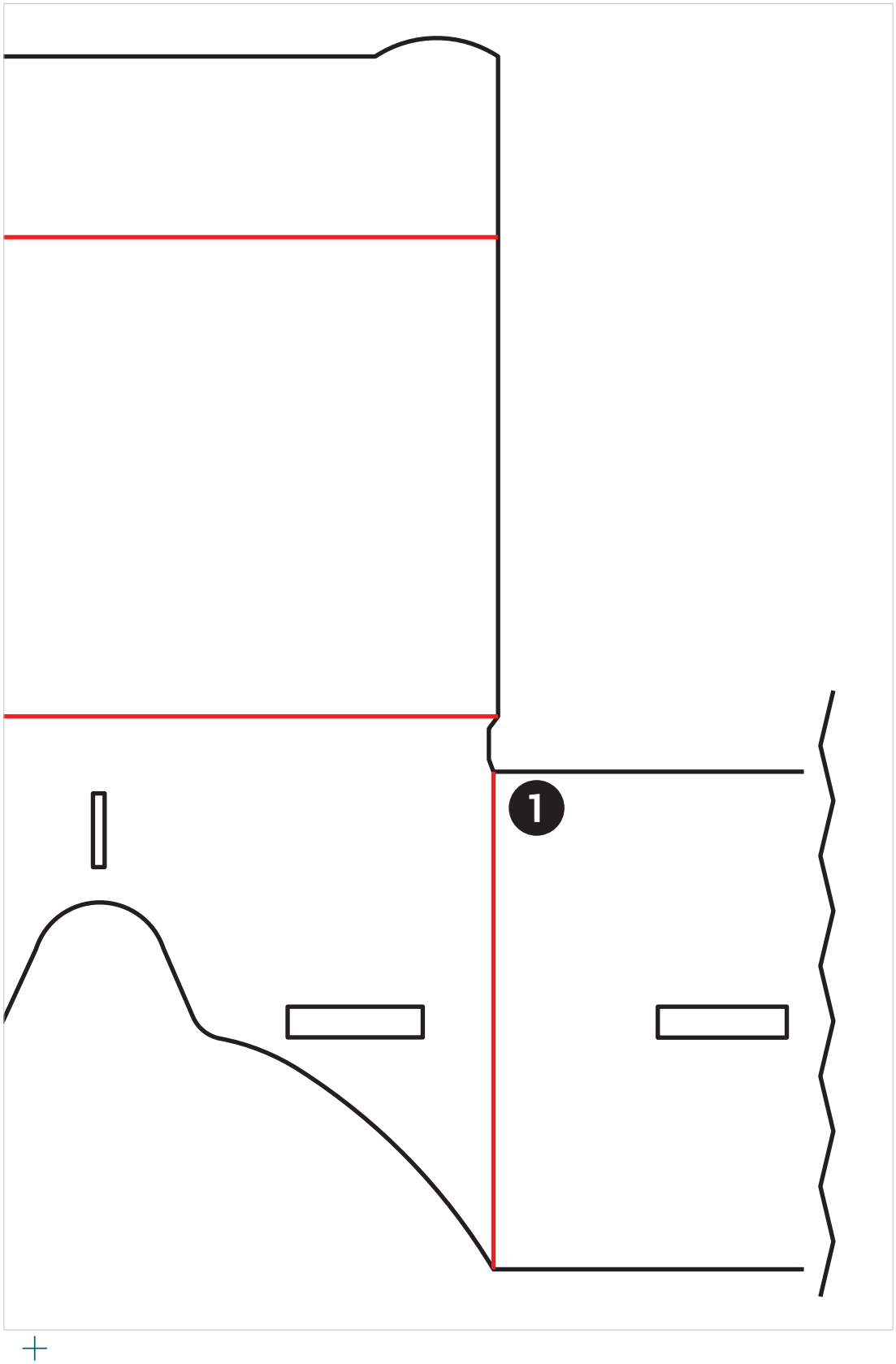
Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap

Template B - right

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



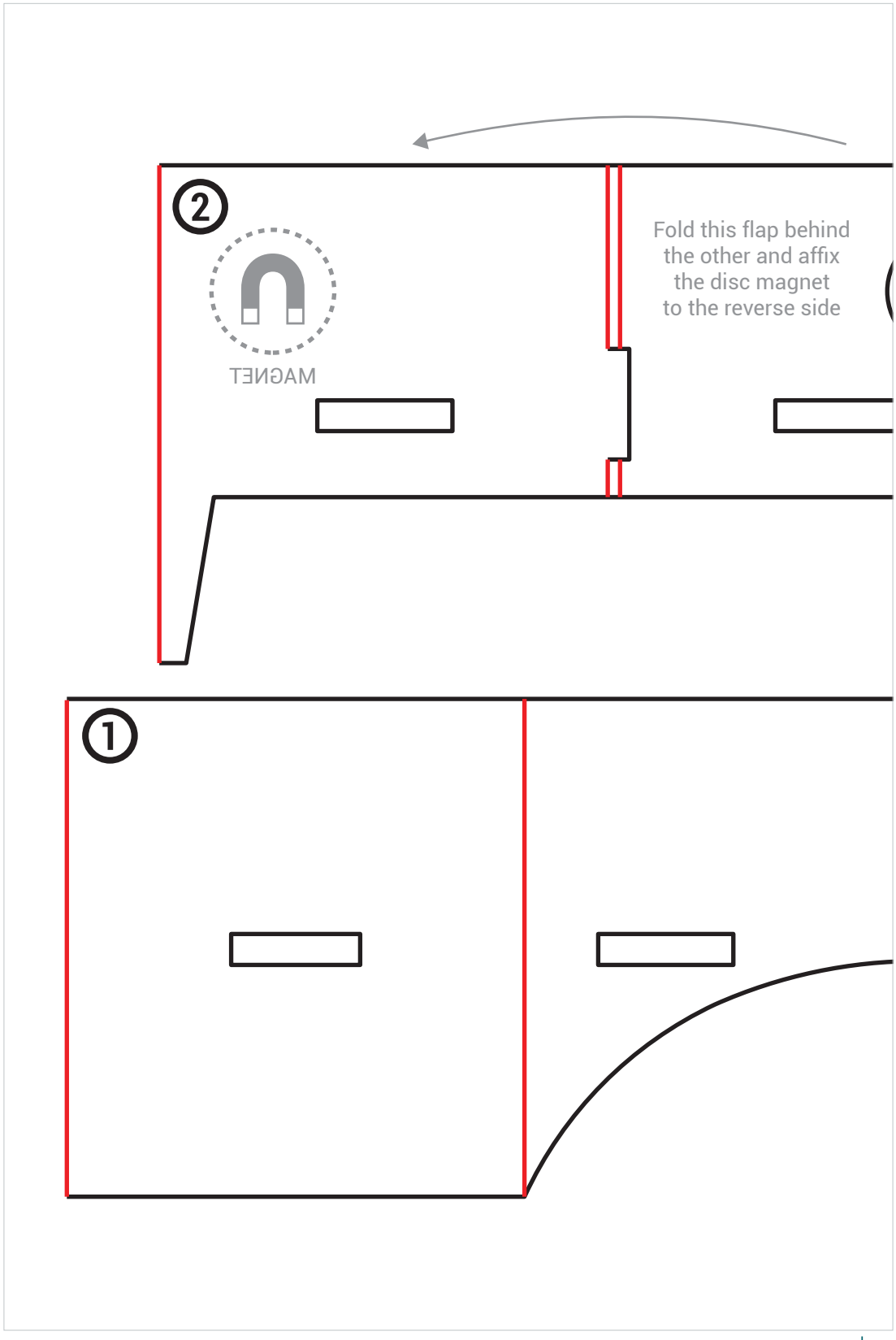
Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap

Template C - left

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



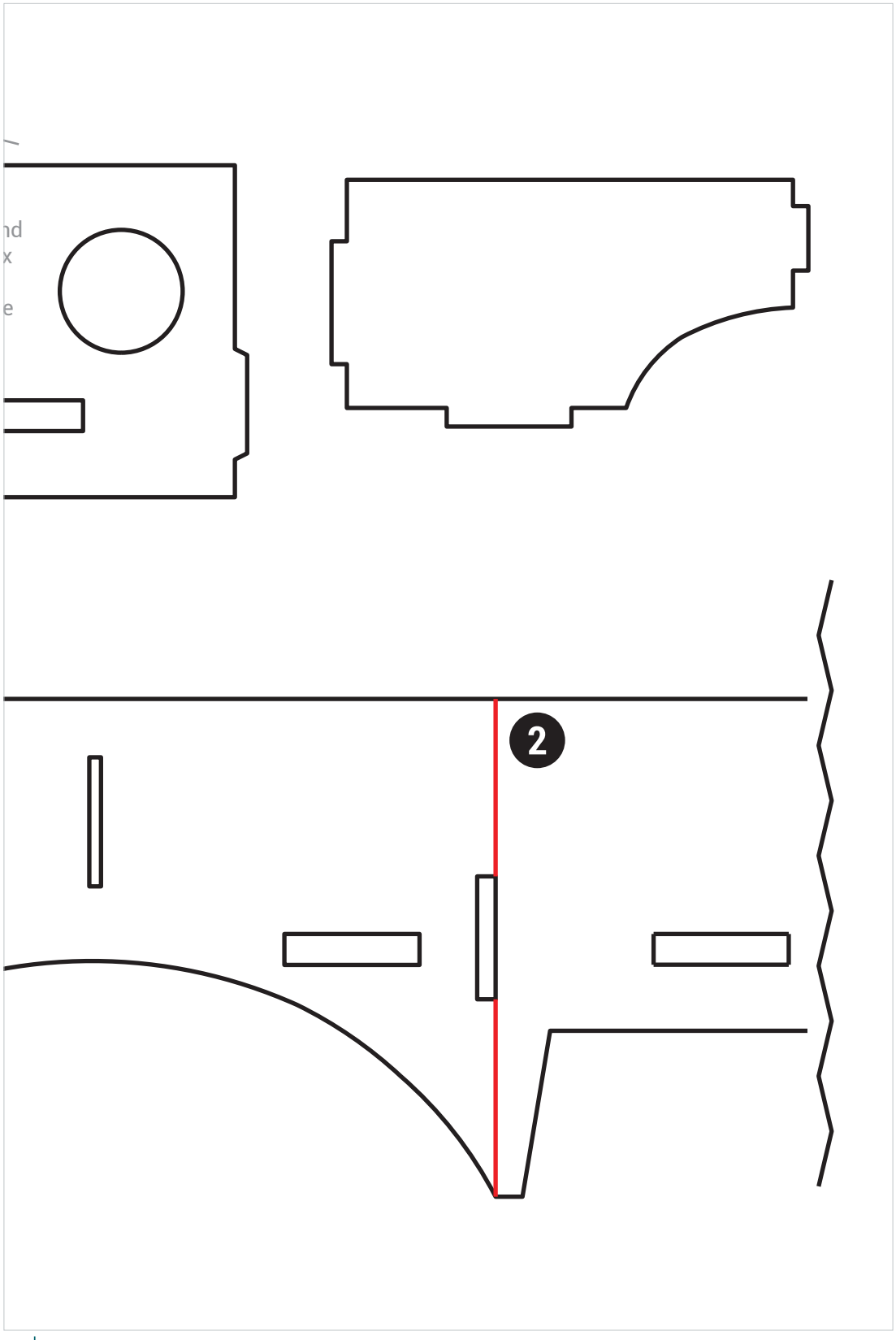
Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap

Template C - right

A4 portrait | 100% / actual size | scale-check ruler: 50 mm



Join to the other half with the same letter. Align both crosses.

This line must measure exactly 5 cm.

Black = cut | Red = fold | 10 mm overlap